VPN Log Investigation in Elastic

Using Kibana to find a brute force attack and a leaver's account still in use

Overview

In this project I loaded a month of VPN logs into Elasticsearch and investigated them in Kibana. I used the Discover tab, the time histogram and the fields pane to spot problems, KQL to dig into them, and Lens to build visualisations. I finished by putting it all on one dashboard so the team could watch VPN activity at a glance.

Scenario

Northvale Logistics is a fictional UK company. Staff connect through a VPN, and every connection, failed login and disconnect is logged. HR also told the security team that p.lloyd left the company on 15 September. I was asked to review all VPN activity for September 2026.

The company and its logs are made up. I used AI to help write a Python script that generates the month of logs, with normal staff activity and a few problems hidden inside. This gave me realistic data to practise on.

```
{ "@timestamp": "2026-09-08T09:14:02+01:00", "UserName": "m.clarke",
  "Source_ip": "198.51.100.22", "Source_Country": "United Kingdom",
  "Source_City": "Swindon", "action": "connect", "Company": "Northvale" }
```

The action is connect, teardown (when a session ends) or failed.

Lab Setup

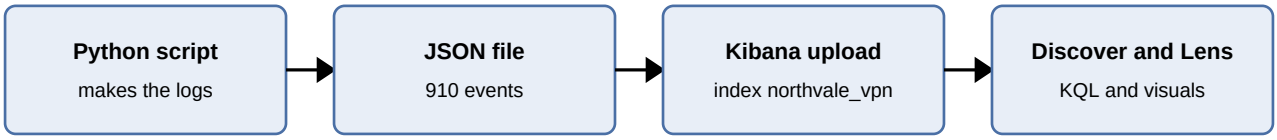


Figure 1. How the VPN data gets into Elastic

- Elasticsearch and Kibana running on my own machine.
- The log file loaded with Kibana's Upload a file tool, into an index called `northvale_vpn`.
- A data view created for the index, using `@timestamp` as the time field.
- The time filter set to 1 September to 1 October 2026 for every search.

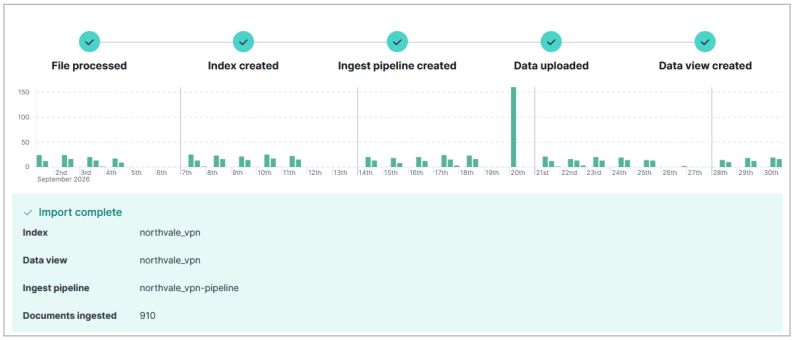


Figure 2. The upload finished with 910 documents in the northvale_vpn index

Step 1: First Look in Discover

With no search at all, Discover showed 910 hits for the month. The histogram at the top shows how many events happened each day. Weekdays all look similar and weekends are almost empty, except for one big spike on Sunday 20 September.



Figure 3. Discover showing all 910 events for September, with the spike on the 20th

A spike on a Sunday is odd for a company that works weekdays, so that was my first lead. There is also a tiny bar on Saturday 26th, which I came back to later.

Step 2: The Sunday Spike

I clicked the spike to zoom in on 20 September. It held 160 events, all between 03:02 and 03:29 in the morning. In the fields pane, the top values for Source_ip showed one IP made 100% of them.

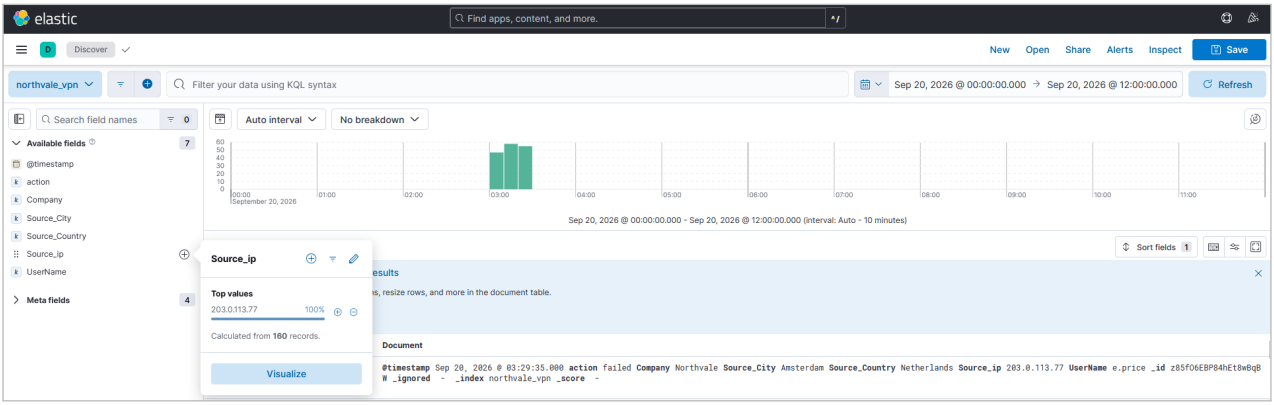


Figure 4. On 20 September, every event came from 203.0.113.77

Next I searched that IP across the whole month with a field based KQL search.

```
Source_ip : "203.0.113.77"
```

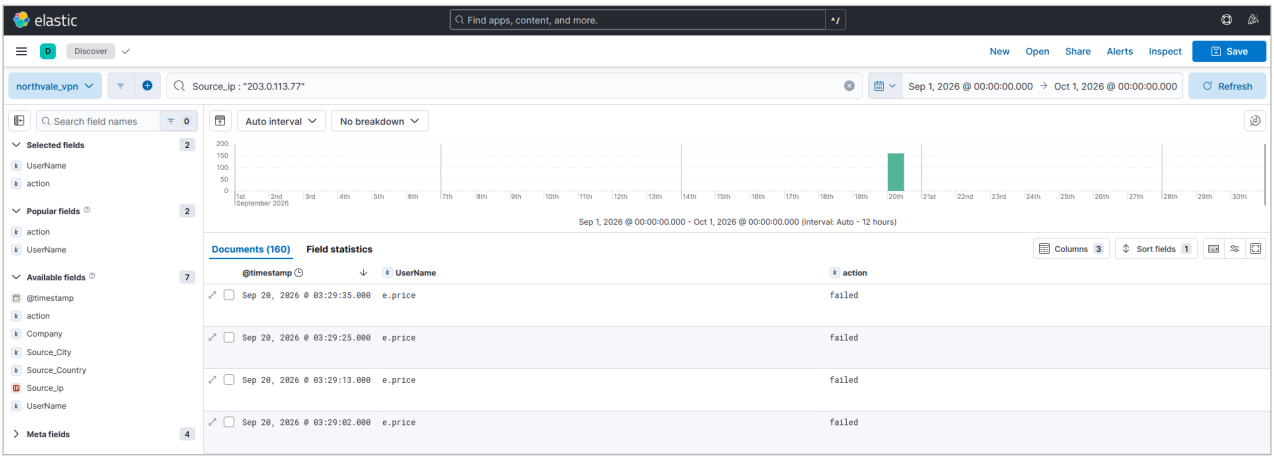


Figure 5. All 160 events from 203.0.113.77, with UserName and action added as columns

The IP only appeared on the 20th, and every event was a failed login to e.price. To check whether any of them got in, I used NOT to hide the failures.

```
Source_ip : "203.0.113.77" and not action : "failed"
```

This returned 0 hits. The IP never logged in successfully. So this was a brute force attack: 160 password guesses against e.price in under 30 minutes, at 3am on a Sunday, and it failed.

Step 3: The Leaver

HR said p.lloyd left on 15 September, so their account should not be used after that. I set the time filter to 15 September to 1 October and searched for the account.

```
UserName : "p.lloyd"
```

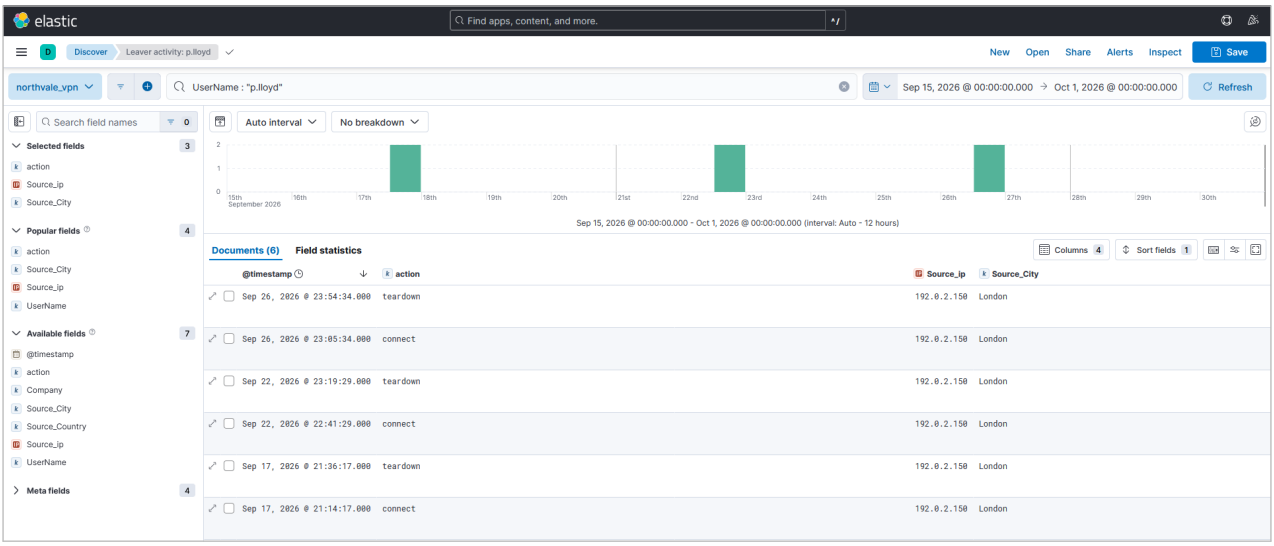


Figure 6. p.lloyd activity after their leaving date, saved as a search for the dashboard

The account connected 3 times after p.lloyd left. All 3 were late at night, one of them on a Saturday, from an IP and city the account never used before. Before leaving, p.lloyd logged in during working hours from their usual IP. Either the account was never disabled and someone is using it, or p.lloyd still has access they should not have. Both are serious. I saved this search as Leaver activity:

p.lloyd so it could go on the dashboard.

Step 4: Checking Logins from Abroad

Northvale staff are based in the UK and Ireland, so I looked for any successful connections from anywhere else. This uses AND, NOT and OR together.

```
action : "connect" and not Source_Country : ("United Kingdom" or "Ireland")
```

This returned 4 hits, all from m.devries in Rotterdam between 7 and 10 September. Each day had a normal working hours session and no failed logins. That pattern fits a work trip, not an attack. I noted it as something to confirm with the user's manager, but not an incident.

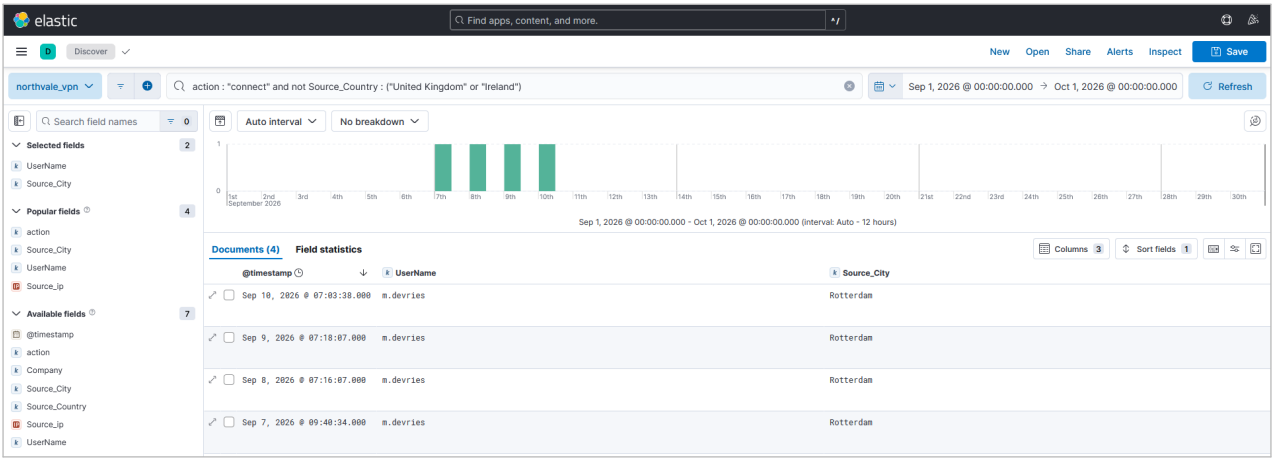


Figure 7. The only connections from outside the UK and Ireland, all m.devries in Rotterdam

Step 5: Visualisations

Failed Logins by User

I filtered to `action : "failed"` (174 hits) and built a Lens table with the top values of UserName and Source_ip.

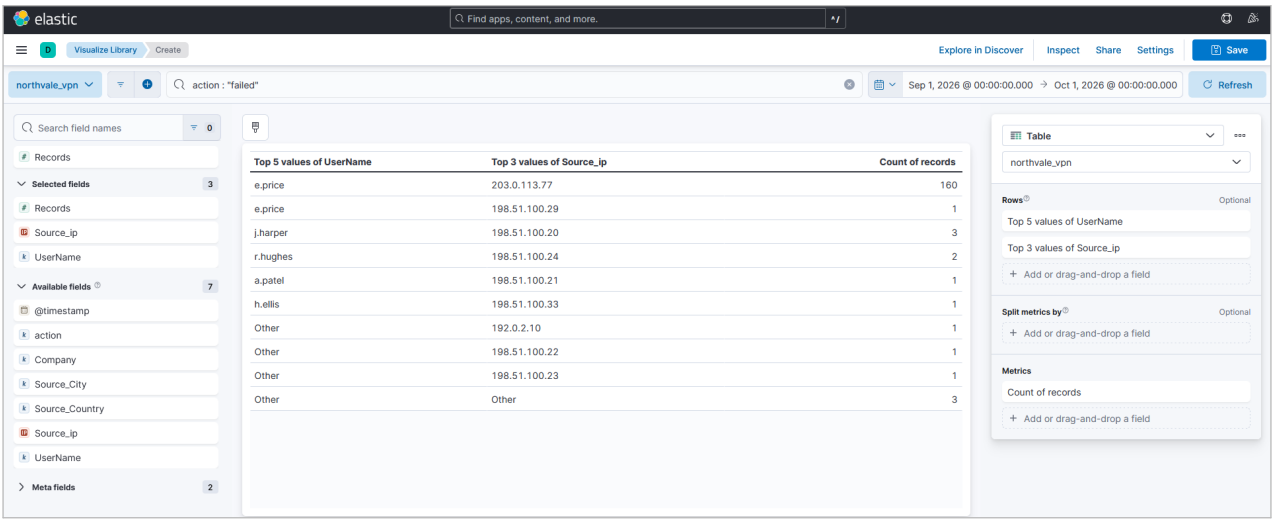


Figure 8. Failed logins by user and IP, saved as Failed logins by user

Apart from the attack, failed logins are rare, 1 to 3 per user across the whole month. That is just people mistyping their password. e.price also has one failed login from their own IP, 198.51.100.29, which looks like a normal typo and shows the real user was not behind the attack. The Other rows are Lens grouping the users outside the top 5.

Connections by Country

I filtered to `action : "connect"` and built a donut chart of Source_Country. It shows 333 connections from the UK, 31 from Ireland and 4 from the Netherlands. I filtered to successful connections on purpose. Without that filter the Netherlands looks huge, because of the 160 failed attack attempts.

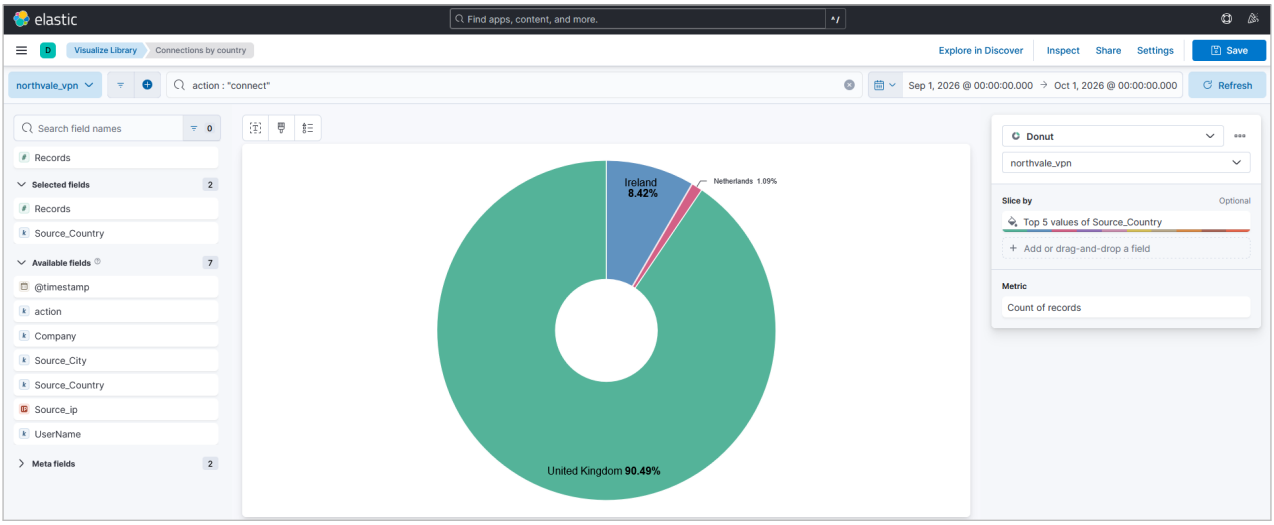


Figure 9. Successful connections by country, saved as Connections by country

Step 6: Dashboard

I added my two saved visualisations and the p.lloyd search to one dashboard called Northvale VPN Overview. I also built a bar chart of events over time straight from the dashboard. An analyst can open it each morning and spot a spike or an odd login in seconds, without writing any queries.

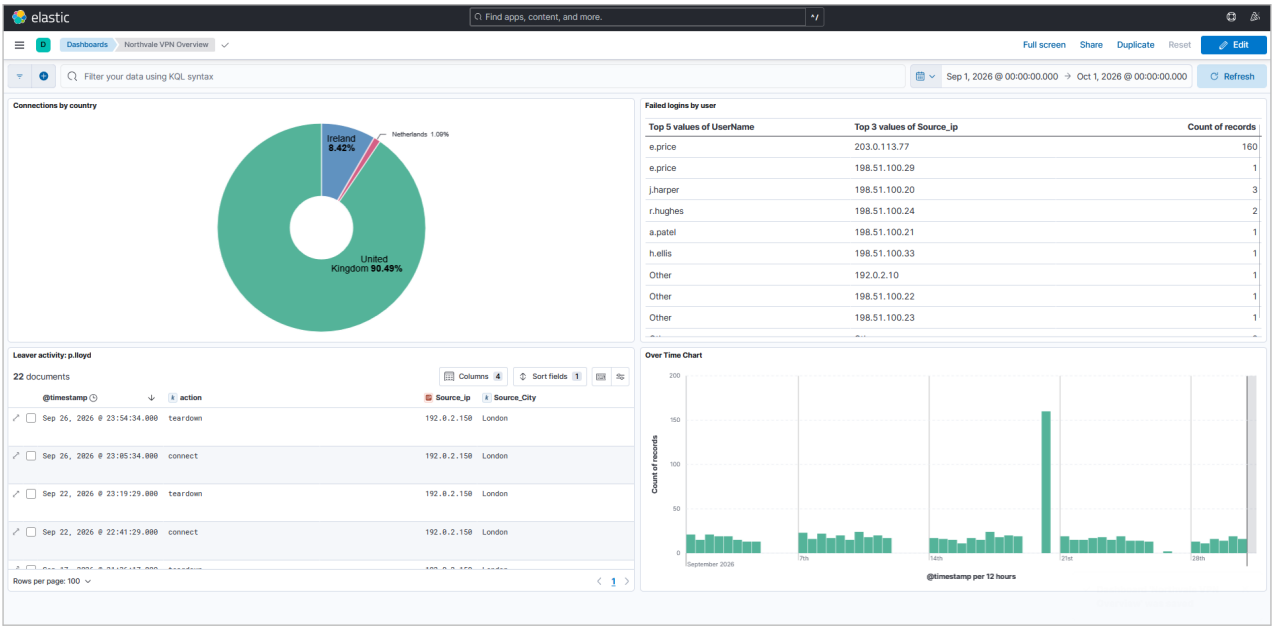


Figure 10. The Northvale VPN Overview dashboard

A dashboard uses one time range for every panel, so the p.lloyd panel shows all 22 of their events for the month, not just the ones after they left. The late night London rows still stand out against their normal daytime logins.

Findings

Finding	Evidence	Verdict	MITRE ATT&CK
Brute force on e.price	160 failed logins from 203.0.113.77 in 28 minutes, none successful	True Positive (failed attack)	T1110.001 Password Guessing
Leaver account in use	p.lloyd connected 3 times after leaving, late at night, from a new IP	True Positive	T1078 Valid Accounts
Logins from the Netherlands	m.devries, 4 normal working hours sessions	False Positive (work trip)	None

Recommended Actions

- Disable the p.lloyd account now and check what it accessed on the VPN.
- Review the leavers process with HR, since the account was not disabled on the leaving date.
- Block 203.0.113.77 and tell e.price their account was targeted.
- Turn on multi-factor authentication and account lockout for VPN logins.
- Confirm m.devries's trip with their manager to close it off.

Splunk and Elastic Compared

I did a similar investigation in Splunk, so here is how the same searches look in each tool.

Task	Splunk (SPL)	Elastic (KQL)
Search one IP	`Source_ip="203.0.113.77"`	`Source_ip : "203.0.113.77"`
Exclude a value	`NOT action="failed"`	`not action : "failed"`
Count by field	`| stats count by UserName`	Fields pane or a Lens table
Charts	Visualization tab	Lens
Saved views	Reports and dashboards	Saved searches and dashboards

SPL felt stronger for counting and doing maths on the data in the search itself. KQL is simpler and only filters, so the counting happens in the fields pane and Lens. Kibana made it quicker to spot things by eye, like the spike in the histogram.

What I Learned

- The time filter matters. If it does not cover the right dates, the data looks like it is missing.
- The histogram is a quick way to spot spikes before writing any query.
- NOT is just as useful as AND, for example to prove an attack never succeeded.
- Context changes the verdict. A login from abroad was fine, but a login from a leaver was not.
- Filtering before visualising matters, or one attack can skew the whole chart.

Skills Demonstrated

- Loading data into Elasticsearch and creating data views
- Kibana Discover, time filter and fields pane
- KQL searching with field, AND, OR and NOT queries
- Lens visualisations and dashboards
- Detecting brute force and leaver account misuse
- Separating true positives from false positives
- MITRE ATT&CK mapping